Average rate of change

1

a $37^\circ\ \mathrm{C}$

b $1.3^\circ\ \mathrm{C/min}$

c $\dfrac{dT}{dt} = 1.4 - 0.04t$

d $1.2^\circ\ \mathrm{C/min}$

e $t = 35$

2

a $\$54$

b $\$8.33$ per year

c $\dfrac{dP}{dx} = x^2 - 16x + 63$

d $\$15$ per year

e $x = 7, x = 9$

3

a $A'(t) = \dfrac{8}{5} + \dfrac{4t}{25}$

b cm^2/h

c $2.8\ \mathrm{cm}^2/\mathrm{h}$

d $A'(t) = 2.56$

4

a $-6000v^{-2}$

b Beats per minute per mL

c 75

d -0.75 bpm/mL

e -0.9375 bpm/mL

Rates of change

5 $\dfrac{dR}{dt} = \dfrac{1}{17} + \dfrac{t}{54}$

6 $\dfrac{dV}{dP} = -kP^{-2}$

7

a $V' = -24x^5 - 25x^4 + 1000x^3$

b $-120\,393$

8

a $\dfrac{dR}{dp} = -6p + 45$

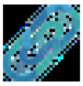
b 21

c -3

d $\$4$, because they will make a revenue of $\$21$, instead of losing $\$3$.

9

a $\dfrac{dy}{dx} = 25x^{-\frac{1}{2}} + \dfrac{40}{3}x^{-\frac{2}{3}} - \dfrac{1}{4}x$



b $\dfrac{dy}{dx} = 7\ \mathrm{kWh/h}$

c Increase

10

a $L' = \dfrac{1}{1000}\left(4t^3 - 24t^2 + 850t - 1300\right)$

b $L' = \dfrac{41\,443}{500}$

c Going in

11

a $\dfrac{3n^2}{2} - 410$

b $17\,740\ \mathrm{m}^2$ per animal

c $204\,940\ \mathrm{m}^2$ per animal

d No, becauseas as the population increases, the area rate increases.

12

a $\dfrac{dV}{dt} = 1000\left(-3t^2 + 150t - 1875\right)$

b $75\,000\ \mathrm{L/s}$

c $t = 15, t = 35$

13

a $f'(x) = -16\,000x^{-3} - 4000x^{-2}$

b How much the voltage generated changes per million kilometres when the probe is a certain distance from the sun.

c $-0.0064\ \mathrm{mV/Gm}$

14

a $P' = 32x^{-\frac{1}{2}} + \dfrac{80}{3}x^{-\frac{2}{3}}$

b

i 15 mice/day ii 7 mice/day iii 5 mice/day

c The growth is increasing but the rate of increase is slowing down.

15

a $\dfrac{dr}{dt} = 1.5t^{-\frac{2}{3}}$

b r increases more slowly.

c $V = 121.5\pi t$

d $\dfrac{dV}{dt} = 121.5\pi$

e $121.5\pi\ \mathrm{cm}^3/\mathrm{s}$

f The volume increases at a constant rate.

16

a $\dfrac{dE}{dg} = 400g - \dfrac{3g^2}{2}$

b $10\,650$

c $133\dfrac{1}{3}$ units

17

a $V = \dfrac{9\pi t^3}{125}$

b $\dfrac{dV}{dt} = \dfrac{27\pi t^2}{125}$

c $\dfrac{432\pi}{125}$ units^3/s

18

a $\$5423$

b $\$11$ per item

c $\dfrac{dC}{dx} = 7 + 60x^{-\frac{1}{2}}$

d $\$9$ per item

19

a $t + \dfrac{9}{10}$ m^2/day

b $\dfrac{209}{10}$ m^2/day

c $t = 38$

20

a $M' = 3x^2 + 36x + 81$

b g/cm

c 221.43 g